Mathematics - BEL2 - English

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Exercises on Differential Equations, Laplace Transform, and Logistics Engineering Applications

Use $j^2 = -1$ throughout whenever complex roots occur. Use $u(t)$ for the unit step function. Give exact results whenever possible. When relevant, determine poles, time constants, transient and steady-state parts, and interpret the results for the logistics system.

1. Classify the following differential equations by:

   - order,
   - linear or nonlinear,
   - homogeneous or non-homogeneous.

$$\dot{B}(t) + 0.4B(t) = 0$$

$$\ddot{x}(t) + 4\dot{x}(t) + 5x(t) = 2$$

$$\dot{q}(t) + q(t)^2 = 10$$

$$t\,\ddot{T}(t) + 2\dot{T}(t) + T(t) = 0$$

2. For each equation below, state how many initial conditions are needed to determine a unique solution. Give one possible set of initial conditions.

$$\dot{I}(t) + 0.5I(t) = 20$$

$$\ddot{x}(t) + 3\dot{x}(t) + 2x(t) = r(t)$$

$$y^{(3)}(t) + 4y'(t) = 0$$

3. The inventory deviation $e(t)$ in a warehouse is modeled by

$$\dot{e}(t) + 0.25e(t) = 0, \qquad e(0) = 80.$$

Determine $e(t)$ and the long-term value

$$\lim_{t\to\infty} e(t).$$

4. The temperature deviation $T(t)$ in a refrigerated logistics container satisfies

$$\dot{T}(t) + \frac{1}{6}T(t) = 0, \qquad T(0) = 12.$$

Determine:

(a) the solution $T(t)$,

(b) the time constant,

(c) the time at which the deviation has decreased to 5% of its initial value.

5. The order backlog $B(t)$ in a distribution center is described by

$$\dot{B}(t) + 0.5B(t) = 30, \qquad B(0) = 0.$$

Determine:

(a) the complete solution $B(t)$,

(b) the equilibrium value,

(c) the transient part.

6. The deviation $y(t)$ of occupied loading docks from a desired operating point obeys

$$\dot{y}(t) + 2y(t) = 10, \qquad y(0) = 8.$$

Determine $y(t)$. Decide whether the solution approaches the equilibrium from above or from below.

7. The speed error $x(t)$ of a conveyor drive satisfies

$$\ddot{x}(t) + 5\dot{x}(t) + 6x(t) = 0.$$

Determine:

(a) the characteristic equation,

(b) its roots,

(c) the general solution,

(d) the qualitative type of response.

8. The positioning error $x(t)$ of a sorting carriage is modeled by

$$\ddot{x}(t) + 4\dot{x}(t) + 4x(t) = 0, \qquad x(0) = 1, \qquad \dot{x}(0) = -2.$$

Determine the complete solution.

9. The horizontal oscillation $x(t)$ of a suspended pallet on an automated crane satisfies

$$\ddot{x}(t) + 2\dot{x}(t) + 10x(t) = 0, \qquad x(0) = 0, \qquad \dot{x}(0) = 5.$$

Determine:

(a) the characteristic roots,

(b) the real-valued solution,

(c) whether the motion is overdamped, critically damped, underdamped, or undamped.

10. For each of the following equations, determine ω_n, ζ, and classify the response:

(a)
$$\ddot{x}(t) + 10\dot{x}(t) + 9x(t) = 0$$

(b)
$$\ddot{x}(t) + 6\dot{x}(t) + 25x(t) = 0$$

(c)
$$\ddot{x}(t) + 8\dot{x}(t) + 16x(t) = 0$$

11. The position $x(t)$ of a package diverter is modeled by

$$\ddot{x}(t) + 2\dot{x}(t) + 5x(t) = 15, \qquad x(0) = 0, \qquad \dot{x}(0) = 0.$$

Determine:

(a) one particular solution,

(b) the homogeneous solution,

(c) the complete solution,

(d) the final value.

12. Compute the following Laplace transforms:

    (a) $\mathcal{L}\{1\}$,
    (b) $\mathcal{L}\{e^{-4t}\}$,
    (c) $\mathcal{L}\{5u(t)\}$,
    (d) $\mathcal{L}\{\cos 3t\}$,
    (e) $\mathcal{L}\{2\sin 4t\}$,
    (f) $\mathcal{L}\{t\}$.

13. Use linearity and the exponential shift property to compute:

    (a)
    $$\mathcal{L}\left\{4 - 3e^{-2t} + 2\sin t\right\}$$

    (b)
    $$\mathcal{L}\left\{e^{-3t}\cos 2t\right\}$$

14. Let
    $$F(s) = \mathcal{L}\{f(t)\}, \qquad f(0) = 3, \qquad f'(0) = -4.$$
    Determine:
    $$\mathcal{L}\{f'(t)\}, \qquad \mathcal{L}\{f''(t)\}.$$

15. Solve the initial value problem by Laplace transform:
    $$\dot{B}(t) + 2B(t) = 8, \qquad B(0) = 1.$$
    Determine both $B(s)$ and $B(t)$.

16. Solve the initial value problem by Laplace transform:
    $$\ddot{x}(t) + 4x(t) = 0, \qquad x(0) = 2, \qquad \dot{x}(0) = -1.$$
    Determine both $X(s)$ and $x(t)$.

17. Determine the inverse Laplace transforms:

    (a)
    $$\mathcal{L}^{-1}\left\{\frac{1}{s+5}\right\}$$

(b)

$$\mathcal{L}^{-1}\left\{\frac{2}{(s+1)^2}\right\}$$

(c)

$$\mathcal{L}^{-1}\left\{\frac{s+2}{(s+2)^2+9}\right\}$$

(d)

$$\mathcal{L}^{-1}\left\{\frac{6}{(s+2)^2+9}\right\}$$

18. The temperature $T(t)$ in a cold-storage room is modeled by

$$5\dot{T}(t) + T(t) = U(t)$$

with zero initial condition. Determine:

(a) the transfer function

$$G(s) = \frac{T(s)}{U(s)},$$

(b) the pole,

(c) the time constant,

(d) whether the natural response is oscillatory or non-oscillatory.

19. A conveyor positioning actuator has the transfer function

$$G(s) = \frac{s+2}{s^2+6s+13}.$$

Determine:

(a) all zeros,

(b) all poles,

(c) a sketch of the poles and zeros in the s-plane,

(d) the qualitative natural response associated with the poles,

(e) whether the system is stable.

20. The throughput response $y(t)$ of an order-processing unit is modeled by

$$4\dot{y}(t) + y(t) = 6u(t), \qquad y(0) = 0.$$

Use the Laplace transform to determine:

(a) $Y(s)$,

(b) a partial fraction decomposition of $Y(s)$,

(c) $y(t)$,

(d) the final value $y(\infty)$,

(e) the time constant of the system.